How Robust Are LLMs to Vietnamese Dialects?

Anonymous submission

Abstract

Large Language Models (LLMs) are typically evaluated on standard written Vietnamese, yet everyday communication frequently involves regional dialects that preserve meaning but differ in surface form. Existing Vietnamese dialect work largely addresses this issue through dialect-to-standard normalization instead of measuring how the model fails under Vietnamese dialectal inputs. To address this gap, we present the first study that systematically evaluates the robustness of LLMs to Vietnamese dialect variation across multiple tasks, quantifying dialect-induced performance degradation and analyzing failure patterns behind it. We introduce **VialectBench** (Vietnamese Dialects Benchmarking), a controlled benchmark for testing whether model decisions remain stable when task content is expressed in six Vietnamese dialect groups. **VialectBench** contains 400 Standard Vietnamese source instances and 2,400 human-written dialectal rewrites spanning emotion recognition (ER), natural language inference (NLI), question answering (QA), and multiple-choice question answering (MCQA). Dataset evaluation with a fixed reference language model shows that the dialectal rewrites induce a measurable model-relative likelihood shift while remaining nearly equal in length to their Standard counterparts. Across ten instruction-tuned models, dialectal inputs reduce average performance by 2.82%, and no evaluated model is fully dialect-invariant. All four tasks are affected, with QA showing the largest average degradation. Robustness also varies substantially across dialect groups: PNT3 and PNT2 cause the largest average performance drops, at 6.17% and 4.73%, respectively, whereas PNB slightly improves average performance by 0.42%.The Central dialect group (PNT1–PNT4) also yields the highest average harmful-flip rate across all models, at 6.54%. These findings show that strong performance on Standard Vietnamese does not guarantee reliable behavior under meaning-preserving regional variation.

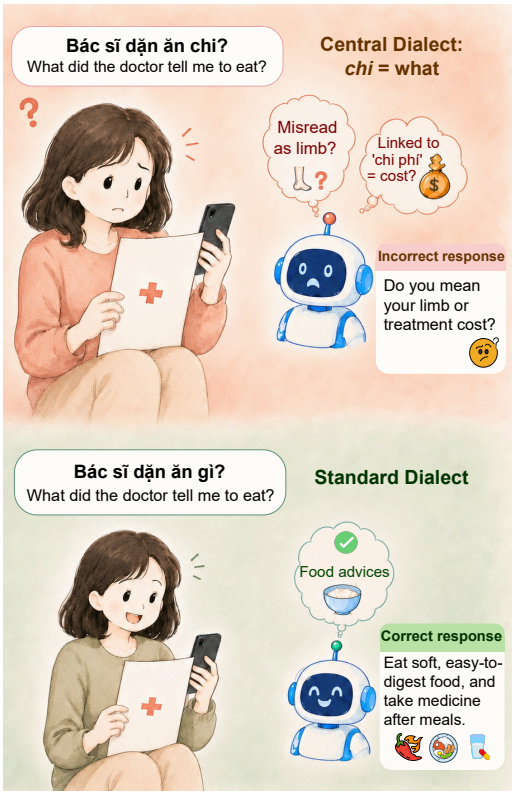


Figure 1: Overview of how Vietnamese dialectal variation affects LLM interpretation and leads to unfaithful responses.

1 Introduction

Large Language Models (LLMs) have become a dominant paradigm in natural language processing (NLP), achieving state-of-the-art performance across a wide range of tasks (Zhao et al. 2026). However, their reliability remains fragile under natural input variation, including changes in wording, style, and linguistic form, which can lead to performance degradation (Zhu et al. 2024). Among these forms of natural variation, dialectal variation is especially important. Unlike synthetic perturbations, dialectal variation reflects language use by real communities, yet it can still shift model predictions and amplify fairness or safety concerns (Hofmann et al. 2024; Faisal et al. 2024). Recent benchmarks have studied this problem in English, Arabic, and multilingual settings (Ziems et al. 2022; Mousi et al. 2025; Faisal et al. 2024; Lin et al. 2025; Gupta et al. 2025), consistently showing that models performing well on standardized language may remain unreliable when processing regional varieties.

This issue is especially important for Vietnamese, where substantial regional variation exists across Northern, Central,

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

and Southern dialects, with further diversity at the provincial level. These varieties differ in lexical choices, pronouns, discourse particles, interrogatives, and function words, many of which can be task-critical for language understanding. Figure 1 illustrates how a meaning-preserving dialectal expression can still lead to model misunderstanding.

Research Gaps and Significance. ❶ Current Vietnamese LLM benchmarks predominantly evaluate models using standardized written Vietnamese, which is closely associated with the Northern variety (Bui et al. 2025; Ho et al. 2020; Huynh, Nguyen, and Nguyen 2025; Luu et al. 2025; Nguyen et al. 2022), rather than controlled regional dialectal variants. Consequently, it remains unclear whether LLMs produce consistent predictions when prompted with meaning-equivalent regional Vietnamese forms rather than standard Vietnamese. ❷ Existing Vietnamese dialect studies have mainly framed regional variation as a dialect-to-standard normalization problem (Le and Luu 2023; Anh Ta, Van Dinh, and Nguyen 2026). Le and Luu (2023) introduce a Central–Northern parallel corpus, while ViDia2Std (Anh Ta, Van Dinh, and Nguyen 2026) provides a nationwide corpus covering all 63 provinces. These resources show that normalizing dialectal inputs can benefit downstream tasks such as sentiment analysis and machine translation. However, they do not directly test whether LLMs themselves are robust to meaning-preserving dialectal inputs. Because they do not provide multiple dialectal variants of the same gold-labeled source instance, these datasets lack the controlled structure required for comparable model decisions across dialects and therefore cannot quantify dialect-induced performance drops, prediction flips, or task-specific failure patterns.

Our work fills this gap by introducing a unified robustness benchmark for evaluating LLMs across emotion recognition (ER), natural language inference (NLI), question answering (QA), and multiple-choice question answering (MCQA). Each standard Vietnamese instance is rewritten into six meaning-preserving regional dialectal variants, while preserving the meaning. Following this design, the standard and dialectal inputs differ primarily in regional linguistic form, allowing observed performance changes to be attributed to dialectal variation. Moreover, the selected tasks require different model capabilities, including reasoning, evidence extraction, and answer selection, allowing us to examine whether dialectal variation affects these abilities differently.

Novelties and Contributions. Our main contributions are summarized as follows:

1. We introduce **VialectBench**, the first end-to-end human-annotated parallel benchmark for evaluating Vietnamese LLM robustness under meaning-preserving dialectal variation, covering six regional dialect groups and four language-understanding tasks.
2. We introduce a controlled framework for evaluating performance and prediction flips across matched Standard Vietnamese and dialectal inputs in four tasks, and benchmark ten open- and closed-source instruction-tuned models across tasks and regions.
3. We illustrate that no evaluated model is fully robust to Vietnamese dialect variation: all four task families are affected, QA exhibits the broadest degradation, and the Central subgroups PNT2 and PNT3 consistently produce the largest performance losses and harmful flips. These results demonstrate that strong Standard Vietnamese performance does not guarantee reliable behavior across regional varieties.

2 Related Work

Dialect Robustness of LLMs. Recent work has begun to benchmark LLM performance under dialectal variation across multiple languages and tasks.

(i) English Dialect Benchmarks. Ziems et al. (2022) introduce Multi-VALUE, a framework for cross-dialectal English NLP using transformations grounded in linguistic features across 50 English varieties. Srirag, Sahoo, and Joshi (2025) introduce M-MD3 to evaluate dialect robustness in conversational understanding, combining naturally occurring US and Indian English dialogues with dialect-transformed and normalized counterparts; their results reveal consistent performance gaps against Indian English in target-word prediction and selection. Lin et al. (2025) present ReDial, which evaluates fairness and robustness on reasoning tasks using over 1,200 parallel Standard English–AAVE query pairs from benchmarks such as HumanEval and GSM8K, demonstrating broad LLM brittleness on AAVE reasoning inputs. Gupta et al. (2025) extend this to five underrepresented English dialects across 12 datasets, showing consistent performance drops on non-standard dialectal inputs.

(ii) Multilingual and Non-English Benchmarks. Faisal et al. (2024) present DialectBench, evaluating 281 language varieties across 40 language clusters and 10 NLP tasks. In Arabic, Mousi et al. (2025) introduce AraDiCE, covering Modern Standard Arabic and three dialects across multiple tasks. Altakrori et al. (2026) adapt MMLU-style question answering into five Arabic dialects.

Vietnamese Dialect studies in NLP. Vietnamese dialect processing has recently attracted increasing attention as regional variation challenges NLP systems trained mainly on standard, Northern-based Vietnamese. Le and Luu (2023) introduce a Central–Northern dialect transfer corpus and show that commercial translation systems struggle with Central dialect expressions, while monolingual Vietnamese models such as BARTpho outperform multilingual models for dialect transfer. Their fine-tuned transfer modules also improve downstream machine translation and text-image retrieval without retraining full systems. Extending this line of work, Anh Ta, Van Dinh, and Nguyen (2026) propose ViDia2Std, a nationwide dialect-to-standard corpus of over 13,000 sentence pairs from all 63 provinces, covering Central, Southern, and non-standard Northern varieties, and show that dialect normalization improves machine translation and sentiment analysis.

Prior work treats Vietnamese dialectal variation mainly as a preprocessing problem, showing that dialect-to-standard normalization can improve downstream systems. However, this does not reveal whether models are inherently robust to dialectal inputs: a model may perform well after normalization yet behave inconsistently on meaning-preserving dialec-

tal variants. *Our work instead focuses on dialectal robustness evaluation, asking whether LLMs produce stable and correct predictions across semantically equivalent Vietnamese dialect forms. This enables controlled analysis of performance drops across tasks, dialect regions, and model families.*

3 Dataset Construction

VialectBench is a manually annotated benchmark comprising 400 standard Vietnamese source instances and 2,400 meaning-preserving dialectal rewrites across six Vietnamese dialect groups. Each source instance is paired with six regional variants. For each source instance, we keep the task label and task-defining evidence fixed while rewriting only the task-bearing input fields: the utterance for ER, the hypothesis for NLI, the question for QA, and both the question and answer options for MCQA. Premises, QA contexts, answer spans, and gold labels are preserved. For MCQA, the semantic identity of each option and the correct answer index are preserved, so that any prediction change reflects dialectal variation rather than a change in the underlying task. Together, these four tasks allow us to evaluate dialectal robustness beyond surface recognition, covering affective understanding, semantic inference, evidence grounding, and answer selection.

We first describe the data sources and sampling strategies in Section 3.1, and then detail the annotation and quality assurance processes in Section 3.2.

3.1 Data Sourcing

In this section, we introduce the selected datasets and their domains. We select these benchmarks to approximate real-world Vietnamese language-use scenarios across diverse domains.

ER – UIT-VSMEC. (Ho et al. 2020) is a Vietnamese social-media emotion recognition dataset consisting of 6,927 human-annotated sentences labeled with six basic emotions and an additional other category.

NLI – ViANLI. (Huynh, Nguyen, and Nguyen 2025) is a Vietnamese adversarial natural language inference dataset containing over 10,000 premise–hypothesis pairs constructed from more than 700 online news articles.

QA – ViQuAD. (Nguyen et al. 2022) is a Vietnamese machine reading comprehension dataset consisting of over 23,000 human-generated question–answer pairs based on 5,109 passages from 174 Vietnamese Wikipedia articles.

MCQA – VIMQA. (Luu et al. 2025) is a Vietnamese multi-hop question-answering dataset containing over 10,000 Wikipedia-based question–answer pairs, which we adapt into a multiple-choice format for answer selection evaluation.

Criterion	Yes	Partial	No
Semantic preservation	72.1%	21.4%	6.5%
Dialect authenticity	58.4%	36.6%	5.0%
Guideline compliance	76.2%	22.3%	1.5%

Table 1: Dual-reviewer evaluation outcomes for all 2,400 dialectal rewrites, rated across three quality criteria.

Rather than relying on random sampling, we adopt a lexicon-guided selection strategy to increase the presence of dialect-sensitive phenomena. Candidate instances are identified using a curated Vietnamese dialect lexicon covering lexical, functional, pronominal, discourse-level, and idiomatic variation. We prioritize examples with richer dialectal trigger coverage while applying task-specific quality constraints, such as valid labels, answerable QA instances, bounded answer length, and well-formed MCQA options.

3.2 Annotation and Quality Assurance

VialectBench follows recent dialect robustness benchmarks that construct parallel inputs through human rewriting or human post-editing rather than relying solely on automatic perturbations (Lin et al. 2025; Mousi et al. 2025; Gupta et al. 2025). For Vietnamese, our design is also informed by prior dialect-to-standard resources (Le and Luu 2023; Anh Ta, Van Dinh, and Nguyen 2026); however, unlike sentence-level normalization corpora, each dialectal rewrite in **VialectBench** is tied to a downstream task label and paired with a standard-control input for direct robustness evaluation.

The six groups cover Northern Vietnamese (PNB), North-Central I (PNT1; Thanh Hoa), North-Central II (PNT2; Nghe An–Ha Tinh), North-Central III (PNT3; Quang Binh–Quang Tri–Thua Thien Hue), South-Central (PNT4; Da Nang–Binh Thuan), and Southern Vietnamese (PNN; Ho Chi Minh City, Southeast Vietnam, and the Mekong Delta). This taxonomy is designed to cover major regional varieties while also distinguishing several Central Vietnamese subregions, where lexical, pronominal, and discourse-marker variation is especially salient.

Figure 2 summarizes the annotation and quality-assurance

Group	Standard PPL	Dialect PPL	PPL ratio	Length ratio
By dialect				
PNB	93.178	82.909	0.926	0.995
PNN	93.178	136.919	1.493	0.984
PNT1	93.178	166.437	1.905	0.997
PNT2	93.178	257.017	2.726	0.991
PNT3	93.178	263.194	**3.334**	0.989
PNT4	93.178	158.199	1.903	0.986
By task				
MCQA	21.998	31.260	1.444	0.995
NLI	132.429	239.129	2.244	0.998
QA	47.171	103.105	**2.385**	0.985
Emotion	171.115	336.290	2.118	0.982
All	93.178	177.446	**2.048**	0.990

Table 2: Dataset quality evaluated with `Qwen2.5-0.5B` as a fixed scorer. Standard and dialect PPL are averaged across sequences, while PPL and length ratios are averaged across matched pairs.

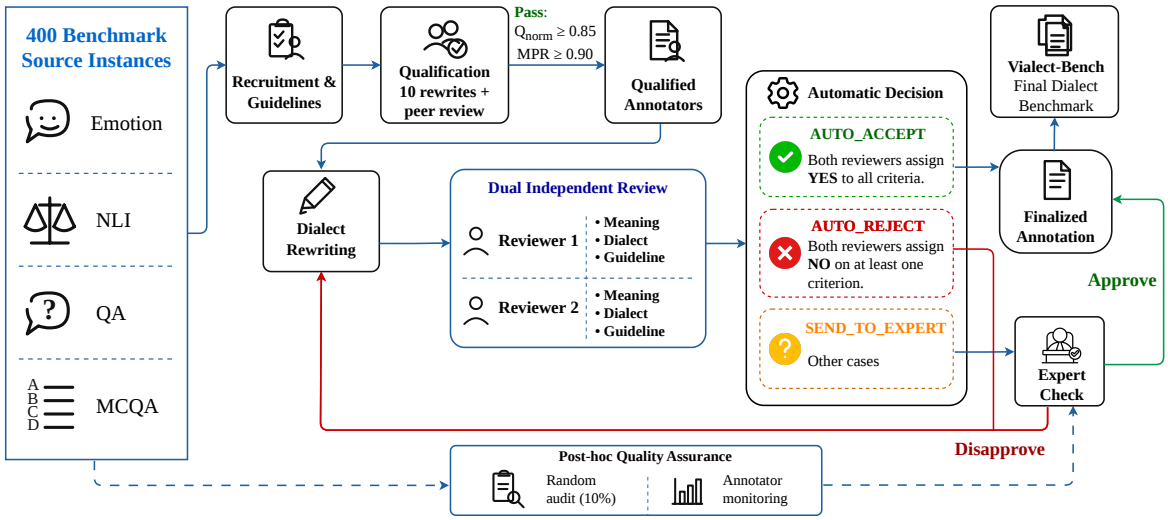


Figure 2: Overview of the annotation and quality-assurance pipeline for **VialectBench**.

workflow. Annotators are recruited from the six dialect regions and self-identified as native speakers or long-term residents of the corresponding region. This follows recent dialect benchmark construction practices that rely on speakers of the target variety to produce or validate dialectal variants (Lin et al. 2025). Before entering the main workflow, annotators complete a qualification stage consisting of dialect rewriting and peer review. Annotators pass qualification when two criteria are met: their combined dialect-naturalness and guideline-compliance score Q_{norm} (the average of reviewer ratings on a 0–1 scale across those two dimensions) is at least 0.85, and their meaning-preservation rate over the qualification rewrites is at least 0.90.

Each submitted rewrite is checked by two independent reviewers along three dimensions: semantic preservation, dialect authenticity, and guideline compliance.

We use dual review and adjudication to reduce individual annotator bias and to support reliable corpus construction, following standard practice in linguistic annotation and inter-coder agreement analysis (Artstein and Poesio 2008).

Table 1 reports the aggregated outcomes from the dual-reviewer evaluation across all 2,400 rewrites. Guideline compliance is the strongest dimension, with 76.2% of rewrites fully passing and only 1.5% failing. Semantic preservation is similarly high at 72.1%, though 6.5% of rewrites were flagged for meaning loss. Dialect authenticity shows the most partial ratings (36.6%), reflecting the difficulty of judging fine-grained regional naturalness.

3.3 Dataset Evaluation

This evaluation characterizes the finalized **VialectBench** before any downstream task evaluation. We use pretrained `Qwen2.5-0.5B` (Qwen et al. 2025) as a frozen causal-language-model scorer to measure how much each dialectal rewrite shifts the model's likelihood relative to its matched Standard Vietnamese control. We score only the task-bearing fields that are rewritten: the utterance for ER, the hypothesis for NLI, the question for QA, and the question together with its answer options for MCQA. For NLI, the Standard comparison is the original hypothesis rather than the unchanged premise. We report the within-pair perplexity ratio as the primary relative measure, along with the whitespace-token length ratio to verify that rewrites remain closely matched in length.

Findings. Table 2 shows a clear likelihood shift in the finalized dataset. Across all 2,400 matched pairs, the mean within-pair PPL ratio is 2.048, while the mean length ratio is 0.990, indicating that the rewrites remain closely matched to their Standard controls in length while exhibiting higher perplexity under the reference scorer.

The shift is strongly dialect-dependent. PNT3 and PNT2 have the largest PPL ratios, at 3.334 and 2.726, respectively, followed by PNT1 (1.905), PNT4 (1.903), and PNN (1.493). PNB is the only group below 1 (0.926), indicating that its rewrites are slightly more probable than their matched Standard forms under this scorer. Across tasks, QA exhibits the largest ratio (2.385), followed by NLI (2.244), ER (2.118), and MCQA (1.444).

These results establish, before downstream evaluation, that **VialectBench** contains a measurable and heterogeneous model-relative surface-form shift. They do not rank dialects by linguistic quality or imply that a larger likelihood shift must cause a larger task-performance drop; Section 4 separately evaluates whether model decisions remain stable under these matched rewrites.

4 Experiments

4.1 Experimental Setting

We evaluate LLM robustness on **VialectBench**, which provides six meaning-preserving dialectal variants for each standard Vietnamese instance across four task families.

Models. We evaluate ten instruction-tuned models: nine open-source models and one proprietary model.❶ The

Task	Variant	Qwen2.5 0.5B	Qwen2.5 3B	Qwen2.5 7B	Llama 3.1 8B	Mistral 7B	Gemma 2 9B	Gemma 3 4B	Vistral 7B	SeaLLM 7B	GPT-4o
MCQA	Standard	68.0	92.0	91.0	87.0	73.0	94.0	89.0	88.0	91.0	94.0
	PNB	63.0	92.0	90.0	87.0	75.0	94.0	90.0	88.0	91.0	94.0
	PNN	63.0	90.0	89.0	85.0	69.0	95.0	88.0	88.0	88.0	94.0
	PNT1	57.0	87.0	88.0	81.0	72.0	93.0	88.0	87.0	85.0	94.0
	PNT2	62.0	87.0	86.0	80.0	70.0	96.0	86.0	85.0	88.0	94.0
	PNT3	57.0	84.0	86.0	81.0	66.0	92.0	88.0	87.0	90.0	93.0
	PNT4	61.0	88.0	88.0	85.0	72.0	95.0	87.0	88.0	88.0	94.0
NLI	Standard	33.0	55.0	64.0	48.0	47.0	63.0	55.0	19.0	46.0	74.0
	PNB	33.0	56.0	66.0	48.0	49.0	65.0	57.0	21.0	48.0	76.0
	PNN	33.0	55.0	59.0	41.0	45.0	60.0	56.0	17.0	45.0	69.0
	PNT1	33.0	54.0	60.0	44.0	51.0	62.0	50.0	19.0	44.0	71.0
	PNT2	33.0	56.0	61.0	47.0	48.0	60.0	54.0	18.0	43.0	70.0
	PNT3	33.0	50.0	59.0	45.0	44.0	57.0	45.0	18.0	42.0	69.0
	PNT4	33.0	55.0	62.0	48.0	50.0	62.0	51.0	18.0	47.0	72.0
QA	Standard	38.9	69.2	74.7	79.4	54.3	84.1	79.6	66.2	74.3	83.2
	PNB	39.5	68.0	73.2	78.6	56.0	82.8	79.9	65.0	74.2	84.6
	PNN	37.6	65.5	71.4	78.5	46.2	83.0	79.8	62.1	72.4	84.1
	PNT1	33.4	65.8	70.2	75.4	48.1	82.8	74.1	63.1	67.0	83.7
	PNT2	18.7	59.1	68.7	75.1	30.5	82.0	73.8	60.3	66.2	86.2
	PNT3	17.1	57.6	67.3	74.5	34.4	80.1	74.5	57.7	61.5	81.6
	PNT4	32.6	60.3	68.0	76.9	37.1	80.0	71.8	60.6	68.1	82.2
Emotion	Standard	24.0	51.0	38.0	43.0	37.0	49.0	43.0	32.0	37.0	60.0
	PNB	23.0	49.0	41.0	47.0	40.0	50.0	45.0	32.0	38.0	56.0
	PNN	24.0	51.0	38.0	44.0	36.0	45.0	41.0	33.0	39.0	56.0
	PNT1	24.0	50.0	38.0	46.0	37.0	45.0	42.0	34.0	39.0	61.0
	PNT2	22.0	38.0	35.0	30.0	33.0	36.0	39.0	31.0	35.0	56.0
	PNT3	20.0	34.0	33.0	30.0	29.0	40.0	41.0	29.0	34.0	60.0
	PNT4	22.0	46.0	44.0	42.0	39.0	43.0	42.0	33.0	37.0	62.0
Mean	Standard	41.0	66.8	66.9	64.3	52.8	72.5	66.6	51.3	62.1	77.8
	PNB	39.6 ↓-1.4	66.3 ↓-0.5	67.5 ↑+0.6	65.1 ↑+0.8	55.0 ↑+2.2	73.0 ↑+0.4	68.0 ↑+1.3	51.5 ↑+0.2	62.8 ↑+0.7	77.7 ↓-0.2
	PNN	39.4 ↓-1.6	65.4 ↓-1.4	64.4 ↓-2.6	62.1 ↓-2.2	49.0 ↓-3.8	70.7 ↓-1.8	66.2 ↓-0.4	50.0 ↓-1.3	61.1 ↓-1.0	75.8 ↓-2.0
	PNT1	36.8 ↓-4.1	64.2 ↓-2.6	64.0 ↓-2.9	61.6 ↓-2.7	52.0 ↓-0.8	70.7 ↓-1.8	63.5 ↓-3.1	50.8 ↓-0.5	58.7 ↓-3.3	77.4 ↓-0.4
	PNT2	33.9 ↓-7.1	60.0 ↓-6.8	62.7 ↓-4.2	58.0 ↓-6.3	45.4 ↓-7.4	68.5 ↓-4.0	63.2 ↓-3.4	48.6 ↓-2.7	58.0 ↓-4.0	76.6 ↓-1.3
	PNT3	31.8 ↓-9.2	56.4 ↓-10.4	61.3 ↓-5.6	57.6 ↓-6.7	43.4 ↓-9.5	67.3 ↓-5.3	62.1 ↓-4.5	47.9 ↓-3.4	56.9 ↓-5.2	75.9 ↓-1.9
	PNT4	37.1 ↓-3.8	62.3 ↓-4.5	65.5 ↓-1.4	63.0 ↓-1.4	49.5 ↓-3.3	70.0 ↓-2.5	62.9 ↓-3.7	49.9 ↓-1.4	60.0 ↓-2.1	77.5 ↓-0.3

Table 3: **Task-by-dialect direct-prompting performance (%)**. Mean rows report the average across the four tasks, with the adjacent values showing the gap from each model's Standard Vietnamese score.

open-source models include three Qwen2.5-Instruct variants (0.5B, 3B, and 7B) (Qwen et al. 2025), Llama-3.1-8B-Instruct (Grattafiori et al. 2024), Mistral-7B-Instruct-v0.3 (Jiang et al. 2023), Gemma-2-9B-IT (Team et al. 2024), and Gemma-3-4B-IT (Team et al. 2025) as general-purpose models, together with Vistral-7B-Chat (Chien Van Nguyen 2023) and SeaLLMs-v3-7B-Chat (Zhang et al. 2025) as Vietnamese- and Southeast Asian-adapted models, respectively.[1] ❷ We additionally evaluate the closed-source model GPT-4o.[2]. This selection enables comparisons across model scales and families, while also examining whether regional language adaptation or stronger general-purpose capabilities improve robustness to Vietnamese dialectal variation.

Implementation. We use direct prompting as the primary inference setting for all models. Each benchmark instance is evaluated in its Standard Vietnamese form and across all corresponding dialectal variants using the same task-specific instruction. Models are required to return a JSON-formatted response, which is subsequently parsed into a task-specific prediction. We use deterministic decoding throughout: for Hugging Face models, we disable sampling by setting `do_sample=False`, whereas GPT-4o is queried with

[1] All open-source models are accessed through Hugging Face.

[2] OpenAI API, https://platform.openai.com/docs/api-reference; we use the fixed `gpt-4o-2024-08-06` API snapshot. Accessed on 5th July 2026.

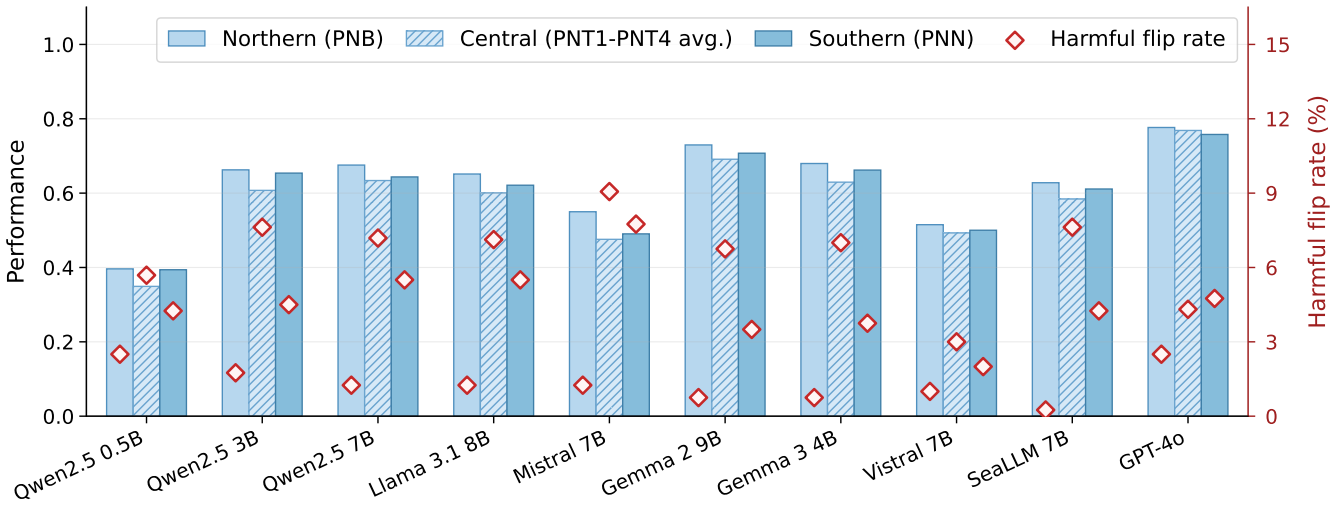


Figure 3: Regional performance (left axis) and harmful-flip rate (right axis) by model.

`temperature=0`. These deterministic decoding settings ensure that differences between Standard Vietnamese and dialectal variants primarily reflect input variation rather than stochasticity during generation.

Evaluation. We evaluate MCQA, NLI, and ER using accuracy, and extractive QA using token-level F1. We additionally report the harmful-flip rate (HFR), defined as the percentage of matched examples answered correctly in Standard Vietnamese but incorrectly in dialectal forms. For QA, correctness in the flip analysis is determined using exact match, while token-level F1 remains the primary performance metric. For MCQA, NLI, and ER, we further measure confidence erosion among examples predicted correctly in both their Standard and dialectal forms. GPT-4o is excluded from this analysis because comparable label probabilities are unavailable.

4.2 Experimental Results

We report task performance on *VialectBench* in Table 3 and Table 4, and summarize the main results of our experiments through these findings.

Dialect brittleness is present across all evaluated models. We define the dialect performance gap as the absolute difference between a model's Standard Vietnamese score and its score on a dialectal variant. As shown in Table 3, PNT3 produces the largest degradation for nine of the ten evaluated models. The most severe cases are Qwen2.5-3B, Mistral-7B, and Qwen2.5-0.5B, whose performance under PNT3 decreases by 10.4%, 9.5%, and 9.2%, respectively. Because these large drops occur across different model families, dialect brittleness is not confined to a particular architecture or model series. At the same time, dialectal variation is not uniformly harmful: PNB occasionally improves performance relative to Standard Vietnamese for some models. This suggests that certain regional reformulations may better match a model's learned lexical or syntactic preferences.

Importantly, absolute capability and dialect stability are distinct. GPT-4o is both the strongest and the most stable model: it achieves the highest Standard Vietnamese score of 77.8% and has an average dialect gap of only approximately 1.0%. In contrast, Vistral-7B also exhibits a relatively small average gap of 1.5%, but its absolute performance remains substantially lower, ranging from 47.9% to 51.5% across dialect groups. Thus, a small dialect gap does not necessarily indicate strong robustness; it may also reflect consistently low performance across both Standard and dialectal inputs. Likewise, occasional improvements under PNB should not be interpreted as dialect invariance, since the same models can still suffer substantial losses under other regional variants.

Task	PNB	PNN	PNT1	PNT2	PNT3	PNT4
MCQA	0.30	1.80	3.50	3.30	4.30	2.10
NLI	-1.50	2.40	1.60	1.40	4.20	0.60
QA	0.21	2.33	4.02	8.31	**9.76**	6.65
Emotion	-0.70	0.70	-0.20	5.90	6.40	0.40
Mean	-0.42	1.81	2.23	4.73	**6.17**	2.44

Table 4: Mean performance gap $S_{\text{Std}} - S_d$ (%) by task and dialect, averaged across ten models. Positive values indicate degradation.

Dialectal variation affects all task families, but QA shows the broadest vulnerability across Central Vietnamese. Tables 3 and 4 show that all four tasks degrade on average, with QA exhibiting the largest mean gap (5.21%), followed by MCQA (2.55%), ER (2.08%), and NLI (1.45%). More importantly, QA is the only task with substantial degradation across all four Central variants: 4.02%, 8.31%, 9.76%, and 6.65% for PNT1–PNT4, respectively. In contrast, ER and NLI are affected more selectively, with their largest losses concentrated in PNT2 and PNT3. Because QA rewrites the question while preserving the Standard Vietnamese context, this pattern suggests that dialectal variation particularly dis-

rupts the alignment between a regional query and its supporting evidence.

NLI has the smallest mean degradation, yet it still drops by 4.20% under PNT3, while improving by 1.50% under PNB. Likewise, ER changes little under PNB, PNN, PNT1, and PNT4, but declines sharply under PNT2 and PNT3 by 5.90% and 6.40%. MCQA follows the same regional pattern, with its largest loss also occurring under PNT3 (4.30%). Figure 3 reinforces this asymmetry: the aggregated Central group generally yields both lower performance and higher harmful-flip rates than PNB and PNN. These results indicate that dialect robustness depends jointly on the task operation and the regional variant, and that averaging across dialects can obscure the concentrated brittleness caused by PNT2 and PNT3.

Central dialects produce both lower performance and more consequential prediction failures. Figure 3 shows a consistent regional asymmetry across model families: the aggregated Central group generally yields the lowest performance and the highest harmful-flip rate, whereas PNB is typically the least disruptive. This indicates that the Central effect is not limited to small changes in aggregate scores; it more often converts predictions that are correct in Standard Vietnamese into incorrect dialectal predictions. The figure also shows that absolute capability and stability must be considered jointly. GPT-4o maintains both high regional performance and relatively low harmful-flip rates, while Vistral-7B appears stable across regions only at a much lower performance level. Conversely, Mistral-7B exhibits both a pronounced Central performance decline and a high harmful-flip rate. Thus, similar average regional gaps can correspond to substantially different levels of practical reliability.

5 Limitations

Although **VialectBench** evaluates controlled written dialect variation, several limitations remain. First, the lexicon-guided sampling strategy intentionally enriches dialect-sensitive phenomena and may therefore over-represent particular lexical or functional patterns relative to naturally occurring Vietnamese. Second, the intrinsic evaluation also relies on a single reference model and tokenizer, so the observed perplexity ordering may be scorer-specific. Finally, we do not evaluate lightweight mitigation strategies, such as dialect-to-Standard normalization or normalization-aware prompting, and therefore cannot determine how much of the observed robustness gap is recoverable without retraining.

6 Conclusion

This paper presented **VialectBench**, a controlled benchmark for evaluating the robustness of LLMs when prompted with six meaning-preserving rewritten Vietnamese regional varieties. The benchmark contains 400 source instances and 2,400 human-written dialectal variants across ER, NLI, QA, and MCQA. Evaluation of ten instruction-tuned models showed that dialectal variation consistently affects performance, and the effect is strongly region-dependent: PNT3 and PNT2 produce the largest average losses, while PNB yields a slight improvement. QA is the most vulnerable task, and the pooled Central group produces the highest harmful-flip rate. Overall, the results show that strong performance on Standard Vietnamese alone is insufficient to establish robustness across regional Vietnamese.

References

Altakrori, M. H.; Habash, N.; Lynn, T.; Samih, Y.; Freihat, A. A.; Chirkunov, K.; AbuOdeh, M.; Florian, R.; Nakov, P.; and Aji, A. F. 2026. DialectalArabicMMLU: Benchmarking Dialectal Capabilities in Arabic and Multilingual Language Models. In Piperidis, S.; Bel, N.; van den Heuvel, H.; Ide, N.; Krek, S.; and Toral, A., eds., *Proceedings of the Fifteenth Language Resources and Evaluation Conference (LREC 2026)*, 3199–3219. Palma, Mallorca, Spain: European Language Resources Association (ELRA).

Anh Ta, K.; Van Dinh, N.; and Nguyen, K. V. 2026. ViDia2Std: A Parallel Corpus and Methods for Low-Resource Vietnamese Dialect-to-Standard Translation. *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(36): 29995–30004.

Artstein, R.; and Poesio, M. 2008. Survey Article: Inter-Coder Agreement for Computational Linguistics. *Computational Linguistics*, 34(4): 555–596.

Bui, C. T.; Son, N. T.; Trang, T. V.; Phung, L. V.; Huy, P. N.; Le, H. A.; Van, Q. H.; Do, P. N.-T.; Truc, V. L. T.; Chau, D. T.; and Nguyen, L.-M. 2025. VMLU Benchmarks: A comprehensive benchmark toolkit for Vietnamese LLMs. In Che, W.; Nabende, J.; Shutova, E.; and Pilehvar, M. T., eds., *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 11495–11515. Vienna, Austria: Association for Computational Linguistics. ISBN 979-8-89176-251-0.

Chien Van Nguyen, Q. N. H. N. B. P. N. P. H. N. P. S. T. N., Thuat Nguyen. 2023. Vistral-7B-Chat - Towards a State-of-the-Art Large Language Model for Vietnamese.

Faisal, F.; Ahia, O.; Srivastava, A.; Ahuja, K.; Chiang, D.; Tsvetkov, Y.; and Anastasopoulos, A. 2024. DIALECT-BENCH: An NLP Benchmark for Dialects, Varieties, and Closely-Related Languages. In Ku, L.-W.; Martins, A.; and Srikumar, V., eds., *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 14412–14454. Bangkok, Thailand: Association for Computational Linguistics.

Grattafiori, A.; Dubey, A.; Jauhri, A.; Pandey, A.; Kadian, A.; Al-Dahle, A.; Letman, A.; Mathur, A.; Schelten, A.; Vaughan, A.; Yang, A.; Fan, A.; Goyal, A.; Hartshorn, A.; Yang, A.; Mitra, A.; and Sravankumar, A. 2024. The Llama 3 Herd of Models. arXiv:2407.21783.

Gupta, A.; Cheung, J.; Meng, P.; Sayyed, S.; Zhu, K.; Liao, A.; and O'Brien, S. 2025. EnDive: A Cross-Dialect Benchmark for Fairness and Performance in Large Language Models. In Christodoulopoulos, C.; Chakraborty, T.; Rose, C.; and Peng, V., eds., *Findings of the Association for Computational Linguistics: EMNLP 2025*, 16830–16855. Suzhou, China: Association for Computational Linguistics. ISBN 979-8-89176-335-7.

Ho, V. A.; Nguyen, D. H.-C.; Nguyen, D. H.; Pham, L. T.-V.; Nguyen, D.-V.; Nguyen, K. V.; and Nguyen, N. L.-T. 2020. Emotion Recognition for Vietnamese Social Media Text. In Nguyen, L.-M.; Phan, X.-H.; Hasida, K.; and Tojo, S., eds., *Computational Linguistics*, 319–333. Singapore: Springer Singapore. ISBN 978-981-15-6168-9.

Hofmann, V.; Kalluri, P. R.; Jurafsky, D.; and King, S. 2024. AI generates covertly racist decisions about people based on their dialect. *Nature*, 633(8028): 147–154.

Jiang, A. Q.; Sablayrolles, A.; Mensch, A.; Bamford, C.; Chaplot, D. S.; de las Casas, D.; Bressand, F.; Lengyel, G.; Lample, G.; Saulnier, L.; Lavaud, L. R.; Lachaux, M.-A.; Stock, P.; Scao, T. L.; Lavril, T.; Wang, T.; Lacroix, T.; and Sayed, W. E. 2023. Mistral 7B. arXiv:2310.06825.

Le, T.; and Luu, A. 2023. A Parallel Corpus for Vietnamese Central-Northern Dialect Text Transfer. In Bouamor, H.; Pino, J.; and Bali, K., eds., *Findings of the Association for Computational Linguistics: EMNLP 2023*, 13839–13855. Singapore: Association for Computational Linguistics.

Lin, F.; Mao, S.; La Malfa, E.; Hofmann, V.; de Wynter, A.; Wang, X.; Chen, S.-Q.; Wooldridge, M. J.; Pierrehumbert, J. B.; and Wei, F. 2025. Assessing Dialect Fairness and Robustness of Large Language Models in Reasoning Tasks. In Che, W.; Nabende, J.; Shutova, E.; and Pilehvar, M. T., eds., *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 6317–6342. Vienna, Austria: Association for Computational Linguistics. ISBN 979-8-89176-251-0.

Luu, S. T.; Hoang, K. T.; Pham, T. Q.; Nguyen, K. V.; and Nguyen, N. L.-T. 2025. ViMMRC 2.0 — Enhancing Machine Reading Comprehension on Vietnamese Literature Text. *International Journal of Asian Language Processing*, 35(03).

Mousi, B.; Durrani, N.; Ahmad, F.; Hasan, M. A.; Hasanain, M.; Kabbani, T.; Dalvi, F.; Chowdhury, S. A.; and Alam, F. 2025. AraDiCE: Benchmarks for Dialectal and Cultural Capabilities in LLMs. In Rambow, O.; Wanner, L.; Apidianaki, M.; Al-Khalifa, H.; Eugenio, B. D.; and Schockaert, S., eds., *Proceedings of the 31st International Conference on Computational Linguistics*, 4186–4218. Abu Dhabi, UAE: Association for Computational Linguistics.

Nguyen, K.; Tran, S. Q.; Nguyen, L. T.; Huynh, T. V.; Luu, S. T.; and Nguyen, N. L.-T. 2022. VLSP 2021-ViMRC Challenge: Vietnamese Machine Reading Comprehension. *VNU Journal of Science: Computer Science and Communication Engineering*, 38(2).

Qwen; :; Yang, A.; Yang, B.; Zhang, B.; Hui, B.; Zheng, B.; Yu, B.; Li, C.; Liu, D.; Huang, F.; Wei, H.; Lin, H.; Yang, J.; Tu, J.; Zhang, J.; Yang, J.; Yang, J.; Zhou, J.; Lin, J.; Dang, K.; Lu, K.; Bao, K.; Yang, K.; Yu, L.; Li, M.; Xue, M.; Zhang, P.; Zhu, Q.; Men, R.; Lin, R.; Li, T.; Tang, T.; Xia, T.; Ren, X.; Ren, X.; Fan, Y.; Su, Y.; Zhang, Y.; Wan, Y.; Liu, Y.; Cui, Z.; Zhang, Z.; and Qiu, Z. 2025. Qwen2.5 Technical Report. arXiv:2412.15115.

Srirag, D.; Sahoo, N. R.; and Joshi, A. 2025. Evaluating Dialect Robustness of Language Models via Conversation Understanding. In *Proceedings of the Second Workshop on Scaling Up Multilingual & Multi-Cultural Evaluation*, 24–38. Abu Dhabi: Association for Computational Linguistics.

Team, G.; Kamath, A.; Ferret, J.; Pathak, S.; Vieillard, N.; Merhej, R.; Perrin, S.; Matejovicova, T.; Ramé, A.; and Rivière, M. 2025. Gemma 3 Technical Report. arXiv:2503.19786.

Team, G.; Riviere, M.; Pathak, S.; Sessa, P. G.; Hardin, C.; Bhupatiraju, S.; Hussenot, L.; Mesnard, T.; Shahriari, B.; Ramé, A.; Ferret, J.; Liu, P.; and Tafti, P. 2024. Gemma 2: Improving Open Language Models at a Practical Size. arXiv:2408.00118.

Van Huynh, T.; Van Nguyen, K.; and Luu-Thuy Nguyen, N. 2026. A new benchmark dataset and mixture-of-experts language models for adversarial natural language inference in Vietnamese. *Expert Systems with Applications*, 306: 130109.

Zhang, W.; Chan, H. P.; Zhao, Y.; Aljunied, M.; Wang, J.; Liu, C.; Deng, Y.; Hu, Z.; Xu, W.; Chia, Y. K.; Li, X.; and Bing, L. 2025. SeaLLMs 3: Open Foundation and Chat Multilingual Large Language Models for Southeast Asian Languages. In Dziri, N.; Ren, S. X.; and Diao, S., eds., *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (System Demonstrations)*, 96–105. Albuquerque, New Mexico: Association for Computational Linguistics. ISBN 979-8-89176-191-9.

Zhao, W. X.; Zhou, K.; Li, J.; Tang, T.; Wang, X.; Hou, Y.; Min, Y.; Zhang, B.; Zhang, J.; Dong, Z.; Du, Y.; Yang, C.; Chen, Y.; Chen, Z.; Jiang, J.; Ren, R.; Li, Y.; Tang, X.; Liu, Z.; Liu, P.; Nie, J.-Y.; and Wen, J.-R. 2026. A Survey of Large Language Models. arXiv:2303.18223.

Zhu, K.; Wang, J.; Zhou, J.; Wang, Z.; Chen, H.; Wang, Y.; Yang, L.; Ye, W.; Zhang, Y.; Gong, N.; and Xie, X. 2024. PromptRobust: Towards Evaluating the Robustness of Large Language Models on Adversarial Prompts. In *Proceedings of the 1st ACM Workshop on Large AI Systems and Models with Privacy and Safety Analysis*, LAMPS '24, 57–68. New York, NY, USA: Association for Computing Machinery. ISBN 9798400712098.

Ziems, C.; Held, W.; Yang, J.; Dhamala, J.; Gupta, R.; and Yang, D. 2022. Multi-VALUE: A Framework for Cross-Dialectal English NLP.